Oracle

Retrained on Homology-Restricted MegaScale

Zero-shot

Supervised, not trained on Megascale

Rectified gain @ k=96 (kcal/mol)

Predicted most stabilizing per domain

0

1

10

6.50 6.49 13.25 16.65

3.51 3.74 9.00 10.17 3.69 3.81 9.51 11.18 2.08 3.00 6.53 8.65 2.80 2.82 8.58 9.57 2.97 3.36 8.89 9.63

2.57 3.59 3.26 4.34 3.08 4.55 4.15 3.89 2.87 3.41 4.69 4.07

2.60 1.66 6.74 5.45 2.07 2.25 6.42 6.60

12.97 17.18 26.88 38.97

12.10 17.03 28.09 38.78 11.54 16.31 27.70 40.45 10.02 17.91 25.29 42.78 11.24 15.17 29.63 39.96 11.46 14.82 30.72 39.00

20.73 30.67 13.91 30.70 23.24 32.18 16.21 24.36 21.85 27.85 21.30 25.00

19.48 10.91 33.13 32.42 11.97 17.06 28.00 38.97

Top 96 per domain

ESM-MSR (p0) WT head

ESM-MSR (p0) combined

MutateEverything st1

ThermoMPNN

SPURS

ESM3-small-open unmasked

ProteinMPNN

ProSST

GeoDDG-3D

Rosetta

Δ hydrophobicity ≤ 0 & relative ASA ≤ 0.5

Δ hydrophobicity > 0 & relative ASA ≤ 0.5

Δ hydrophobicity ≤ 0 & relative ASA > 0.5

Δ hydrophobicity > 0 & relative ASA > 0.5

Top 96 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,295 single mutants); gain = sum of max(ddG, 0) over every pick, whatever its predicted sign; boxes: interquartile range with darker colored median line; dots: domains; dark bars & annotations: mean over domains; dashed lines: oracle mean.